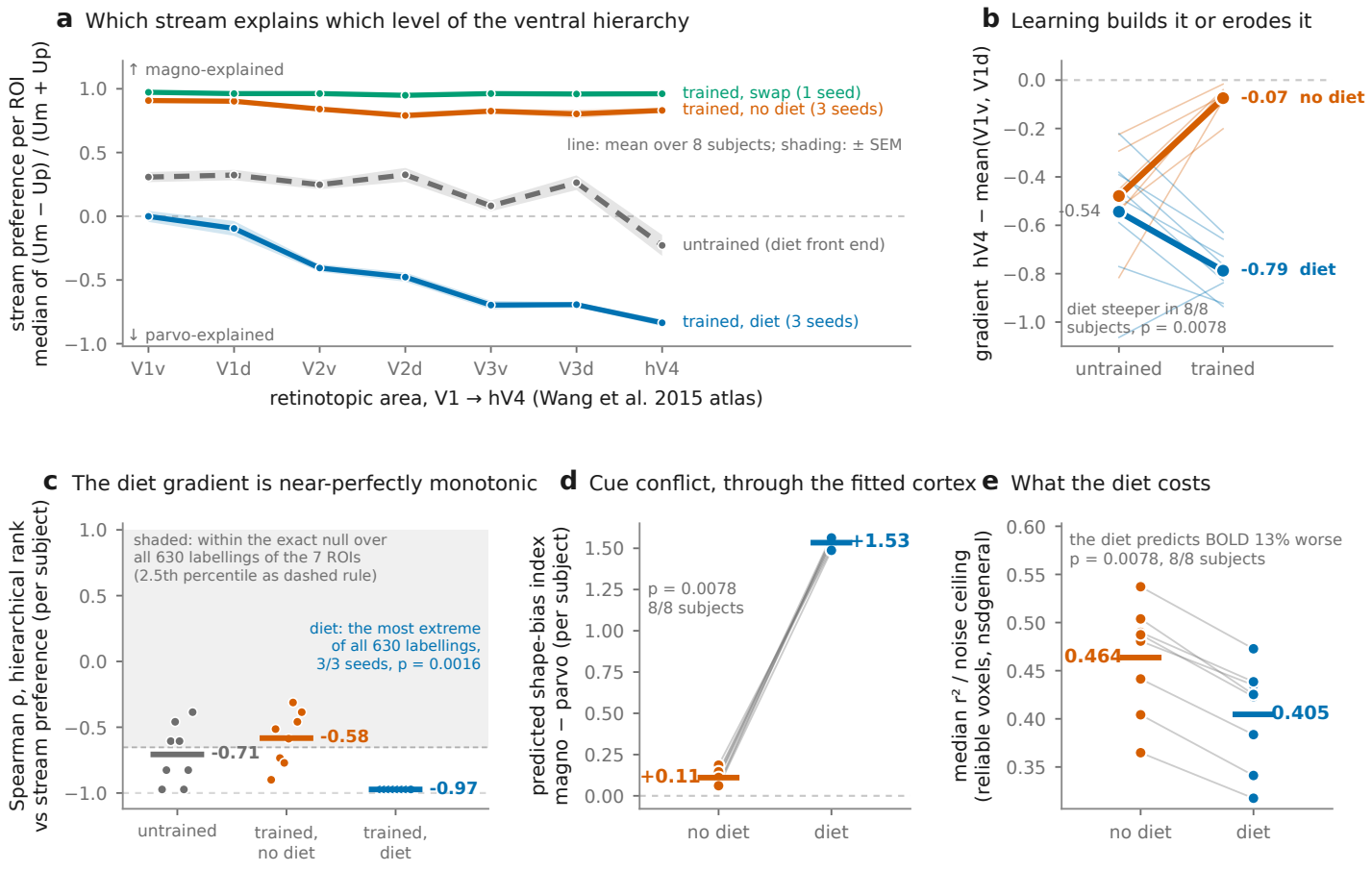
a Which stream explains which level of the ventral hierarchy
↑ magno-explained
↓ parvo-explained
stream preference per ROI
median of (Um − Up) / (Um + Up)
1.0
0.5
0.0
−0.5
−1.0
trained, swap (1 seed)
trained, no diet (3 seeds)
line: mean over 8 subjects; shading: ± SEM
untrained (diet front end)
trained, diet (3 seeds)
V1v
V1d
V2v
V2d
V3v
V3d
hV4
retinotopic area, V1 → hV4 (Wang et al. 2015 atlas)
b Learning builds it or erodes it
gradient hV4 − mean(V1v, V1d)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
-0.07 no diet
-0.54
-0.79 diet
diet steeper in 8/8
subjects, p = 0.0078
untrained
trained
c The diet gradient is near-perfectly monotonic
Spearman ρ, hierarchical rank
vs stream preference (per subject)
1.0
0.5
0.0
−0.5
−1.0
shaded: within the exact null over
all 630 labellings of the 7 ROIs
(2.5th percentile as dashed rule)
diet: the most extreme
of all 630 labellings,
3/3 seeds, p = 0.0016
-0.71
-0.58
-0.97
untrained
trained,
no diet
trained,
diet
d Cue conflict, through the fitted cortex
predicted shape-bias index
magno − parvo (per subject)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
p = 0.0078
8/8 subjects
+1.53
+0.11
no diet
diet
e What the diet costs
median r² / noise ceiling
(reliable voxels, nsdgeneral)
0.60
0.55
0.50
0.45
0.40
0.35
the diet predicts BOLD 13% worse
p = 0.0078, 8/8 subjects
0.464
0.405
no diet
diet